Student Number

PEM

2022

Year 11 End of Year Examination

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time - 2 hours
- Write using black pen
- NESA approved calculators may be used
- A NESA reference sheet is provided
- In Questions 13 – 30, show relevant mathematical reasoning and/or calculations

Total Marks – 80

Section I Questions 1 – 12 **12 marks**
Allow about 20 minutes for this section

Section II Questions 13 – 30 **68 marks**
Allow about 1 hour and 40 minutes for this section

Section I (12 marks)

Write your answers on the multiple choice answer sheet.

Allow about 20 minutes for this section.

1 If Ji can pack 36 boxes per hour. How many can she pack in 20 minutes?

(A) 720
(B) 18
(C) 12
(D) 72

2 Choose the correct solution to the following equation.

$$3(x+1) = 2(5-x)$$

(A) 7
(B) 5
(C) -1
(D) 1.4

3 Evaluate the following equation for *D*,

$S = \frac{D}{T}$ by substituting $S = 60$km/h and $T = 3.5$h.

(A) $210\ km$
(B) $17.14\ km$
(C) $0.06\ km$
(D) $2.1\ km$

4 How many tablespoons (15mL) of medicine are in a medicine bottle with capacity of 0.9L?

(A) 6
(B) 13.5
(C) 600
(D) 60

5 The XPT leaves Sydney Central Railway station at 7:40 every weekday morning and arrives at Melbourne Southern Cross station at 18:30.
How many hours is the trip?

(A) 10h10 $mins$
(B) 11h 10 $mins$
(C) 11h 30 $mins$
(D) 10h 50 $mins$

6 A 1L container of almond milk has the following nutritional information

Servings per package: 4	Serving size 250ml (1 cup)	
	Average Quantity per serve	Average Quantity per 100ml
Energy	**Y**kj	154kj
Protein	2.0g	0.8g
Fat	6.1g	**X**g
Carbohydrate	6.9g	2.8g
Sodium	104mg	41mg
Sugar	4.3g	1.7g

The values of X, the number of grams of fat in 100mL and Y, the number of kilojoules in one cup of almond milk should be

(A) $X = 2\ and\ Y = 385$
(B) $X = 2.44\ and\ Y = 385$
(C) $X = 2.44\ and\ Y = 396$
(D) $X = 2\ and\ Y = 396$

7 The interquartile range of the ages of people who attended a car show last Saturday. listed below, is:

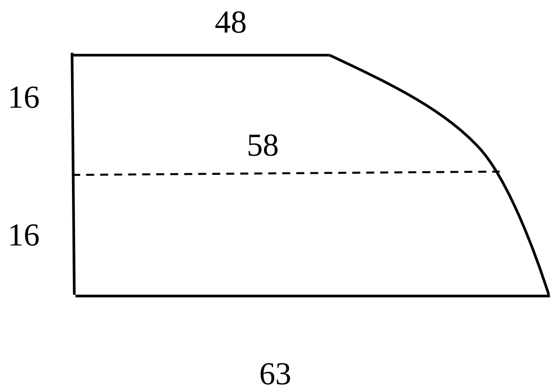
29, 27, 37, 40, 55, 29, 30, 60, 62, 68

(A) 40
(B) 31
(C) 33
(D) 23

8 A motorbike costs $3190.00 including 10% GST.
What is the price before GST was added?

(A) $3500.90
(B) $2871
(C) $3059
(D) $2900

9 Use the trapezoidal rule twice to find the area of the field below



(A) $968m^2$
(B) $848m^2$
(C) $1816m^2$
(D) $1776m^2$

10 Theo is driving 80km/h in outback Australia when a kangaroo jumps across the road in front of him and stops. What is Theo's reaction distance if his reaction time is 1.4 s?

(A) 31.11m
(B) 120 m
(C) 0.03 m
(D) 57.14 m

11 If finance is available on the purchase a vehicle worth \$21400 with \$3000 deposit and monthly repayments of \$380 over 5 years. The total amount paid for the car would be

(A) \$22800
(B) \$24400
(C) \$21400
(D) \$25800

12 Residents in a certain town were surveyed on when they would vote in an upcoming election according to age.
Their responses are recorded below.

	Pre-poll voting	Election day voting	Total
Residents 40 years or over	50	150	
Residents under 40 years age		240	320
Total	130		

How many residents voted?

(A) 520
(B) 390
(C) 560
(D) 450

END OF SECTION I

Student Number

Section II (68 marks)

Attempt Questions 13 - 30

Allow about 1 hour and 40 minutes for this section.

Answer the questions in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided on pages **22** and **23**. If you use this space, clearly indicate which question you are answering.

Write your Student Number at the top of this page.

Please turn over

Question 13. (4 marks)

a) Use the simple interest formula $I = Prn$ to find the number of years it takes for an investment of \$2500 to earn \$525 in interest at 3% pa. **2**

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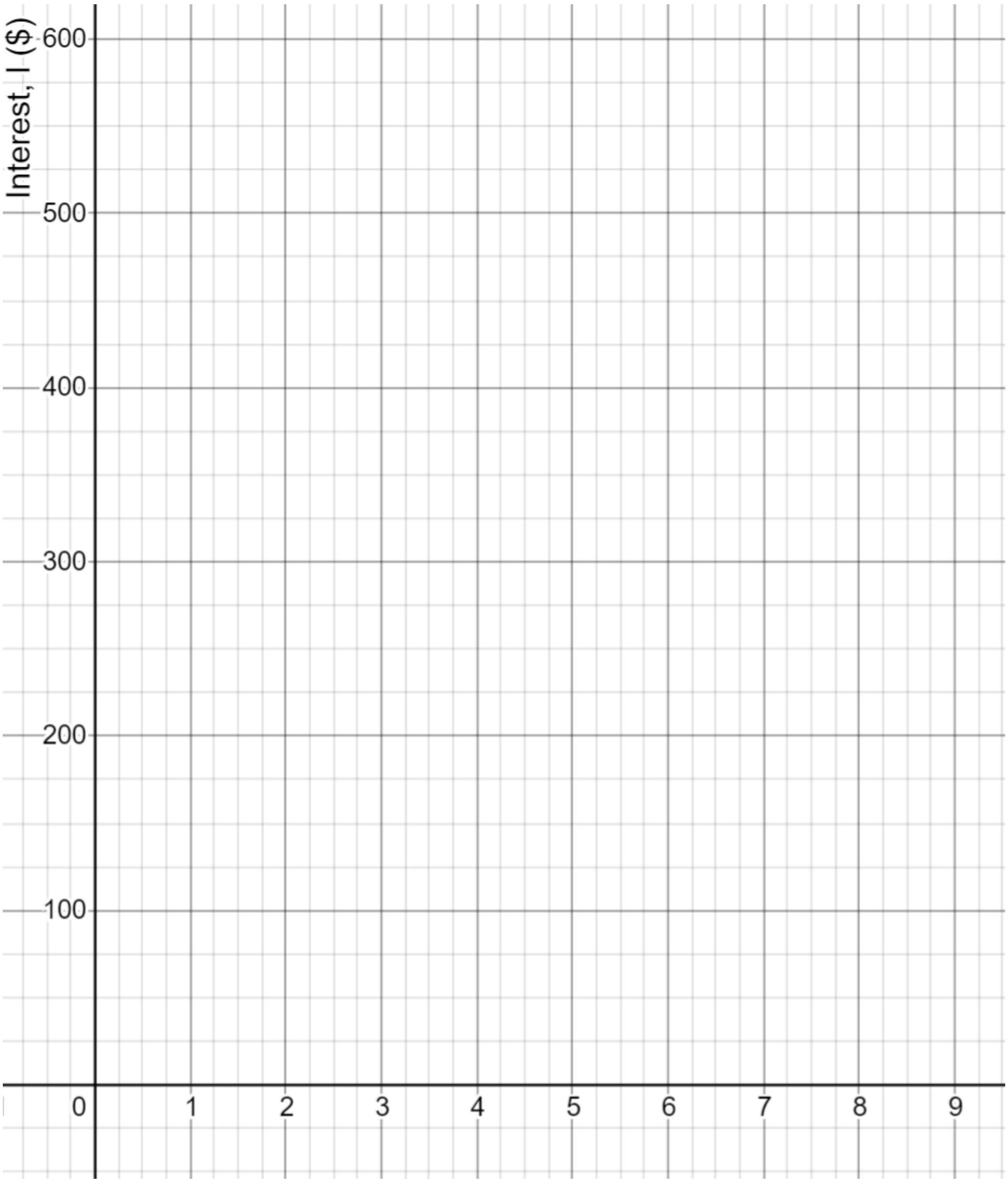
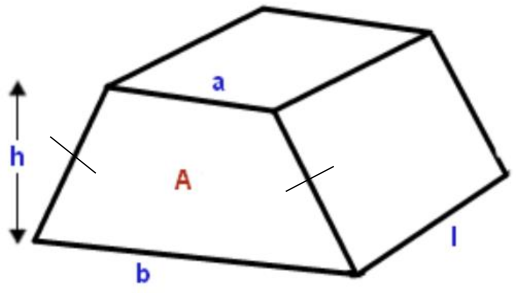
b) Construct a graph to represent the simple interest, *I*, earned in dollars for *n* years, for values of *n* from 0 to 8. **2**

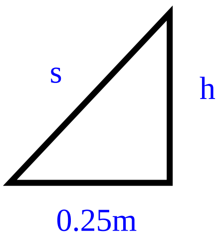


Question 14. (5 marks)

Ali works in a cardboard factory. He has been asked to make a box for a customer as shown below with the following dimensions: $a = 1 \cdot 5$ m, $b = 2$ m, $h = 1 \cdot 4$ m and $l = 3$ m.



a) Ali works out the length of the slant side (s) of the trapezium using this diagram. Use Pythagoras theorem to find the length of s? **2**



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b) For Ali to work out how much cardboard is needed, he needs to know the surface area of the box. Find the surface area.

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c) How much will it cost, to the nearest dollar, if cardboard is 73 cents per square metre?

1

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Question 15. (1 mark)

Perform the following calculation and round your answer to 3 significant figures. **1**

$$(3.2 \times 10^9) \div (2.3 \times 10^7)$$

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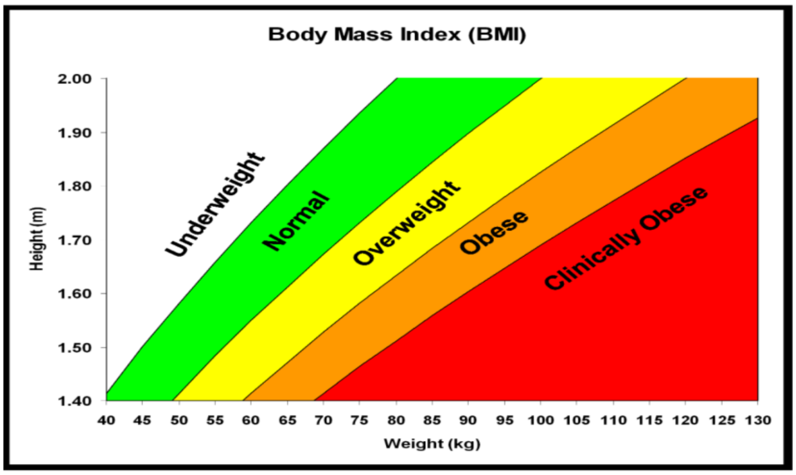
Question 16. (2 marks)

The body mass index (BMI) is used an international measure for health and fitness for adults,

The formula for BMI is $\mathrm{D} = \frac{m}{h^2}$

(where m is the persons mass in kilograms and h is their height in metres)

A 1.56 metre tall 80-year-old woman has been losing weight. The doctor measures her weight as 44.9 kg.



a) Calculate the woman's BMI? **1**

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b) Is the woman now underweight? **1**

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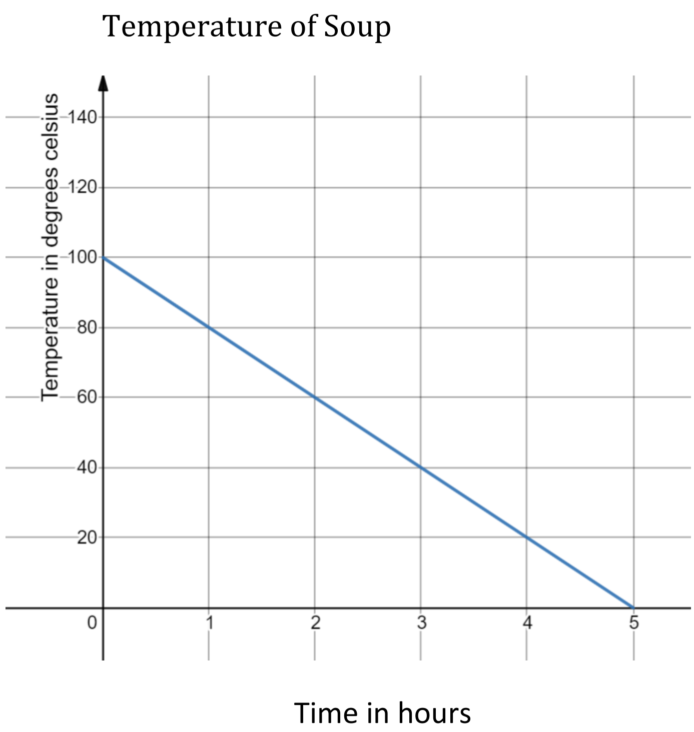
Question 17. (4 marks)

A pot of soup is removed from the stove top and placed in the cool room.

The temperature is measured, straight after placing it in the cool room, as 100℃.

Its temperature decreases according to the graph below.



a) State the gradient and y-intercept of the line. **2**

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b) What is the temperature of the soup after 3 hours? **1**

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c) Approximately how long will it take for the soup to be at a temperature of 50℃ for serving. **1**

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Question 18. (4 marks)

A random sample of students were surveyed at University of Vincenza on the number of times they are red meat that week. The results are shown in the table.

Number of times red meat was eaten this week	Frequency
0	5
1	8
2	12
3	20
4	18
5	15
6	9
7	3

(a) How many students were surveyed? **1**

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(b) What was the most likely times a week that they had eaten red meat? **1**

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(b) Find the relative frequency of a student eating red meat 4 times a week.

2

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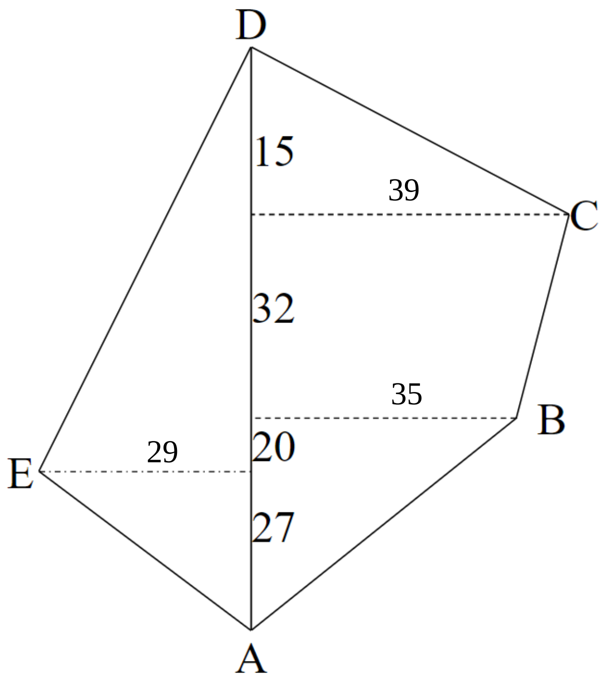
Question 19. (4 marks)

A field surveyor drew the land survey below.



a) Find the length of DC, correct to one decimal place. **2**

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b) Find the area of the field. **2**

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Question 20. (3 marks)

A chef randomly chooses 3 eggs from a dozen to make an omelette, two of the eggs in the carton are rotten.

What is the probability that the chef has chosen two good eggs and one rotten?

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Question 21. (3 marks)

The pool shop owners are charged a peak rate of 25.134 cents per kWh and an off-peak rate of 13.189 cents per kWh (for usage between 10 pm and 7am). **3**

They use their 2800 W pool pump for 3 and a half hours per day, after 10pm, six days a week when the shop closes. How much will their quarterly (12 week) bill be for the pool pump alone?

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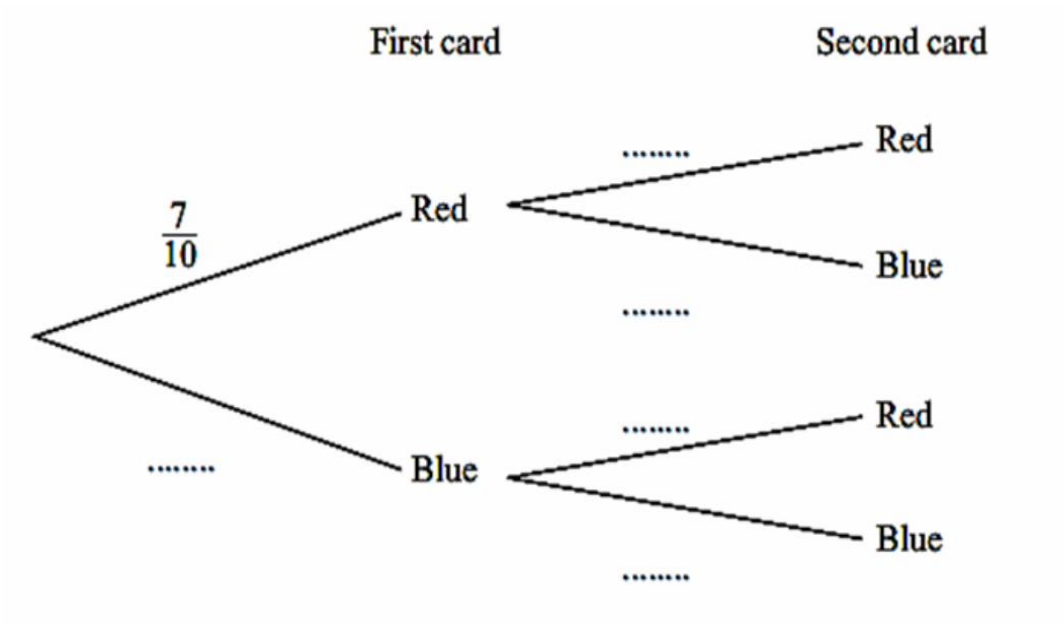
Question 22. (5 marks)

a) Joe holds 7 red cards and 3 Blue cards in his hand.
May chooses one card from his hand and places it in her hand.
She then chooses another card from Joe's hand.
Complete the missing probabilities on the diagram below



3

b) What is the probability that she has choosen 2 cards the same colour? **2**

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Question 23. (3 marks)

An influencer earns 6 cents from her promoter for every like she gets on her site. If a particular week her site is liked 123 409 times,

(a) How much did she earn that week? **1**

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....

(b) She must pay 2.3% of her earnings to the company whose product she is promoting that week, how much does she have to pay the company this week? **1**

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(c) If she worked 52 hours that week preparing, recording and blogging, how much did she earn per hour? **1**

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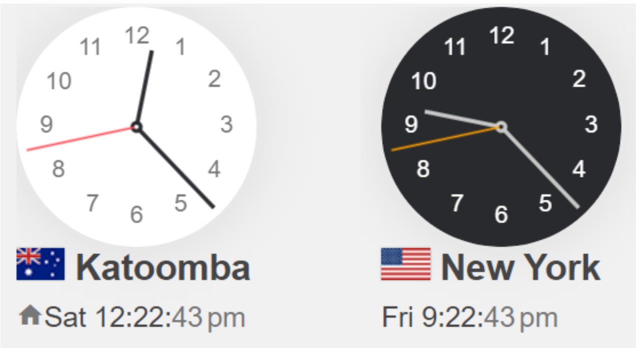
Question 24. (4 marks)



(a) Use the above clocks to find what which city is ahead in time and by how many hours is it ahead of the other? **2**

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(b) What day and time it will be in Katoomba, when it is Friday 12 midnight in New York? **2**

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Question 25. (5 marks)

Minh is single and works part time at 'Woollies' while studying at university. She is paid the casual rate of $21.16 an hour and she worked 20 hours a week for 44 weeks of the year.

(a) How much does she earn before tax for the year? **2**

Taxable income	Tax on this income
0 – $18,200	Nil
$18,201 – $45,000	19 cents for each $1 over $18,200
$45,001 – $120,000	$5,092 plus 32.5 cents for each $1 over $45,000
$120,001 – $180,000	$29,467 plus 37 cents for each $1 over $120,000
$180,001 and over	$51,667 plus 45 cents for each $1 over $180,000

Minh has $250 in deductions for that financial year.

(b) How much will Minh pay in tax? **2**

(c) According to the information from the Medicare website below **1**

Does she have to pay the Medicare levy?

Medicare levy reduction eligibility

In 2020–21, you **do not** have to pay the Medicare levy if:

- you are single, and
- your taxable income is equal to or less than $23,226 ($36,705 for seniors and pensioners entitled to the seniors and pensioners tax offset).

Question 26. (5 marks)

The length of 40 leaves taken from plants of a certain species is shown in the table below.

Length (mm)	Frequency f		
25-29	2		
30-34	4		
35-39	7		
40-44	10		
45-49	8		
50-54	6		
55-59	3		

(a) Find the mean leaf length. **2**

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(b) Find the median. **2**

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(c) Find the modal class. **1**

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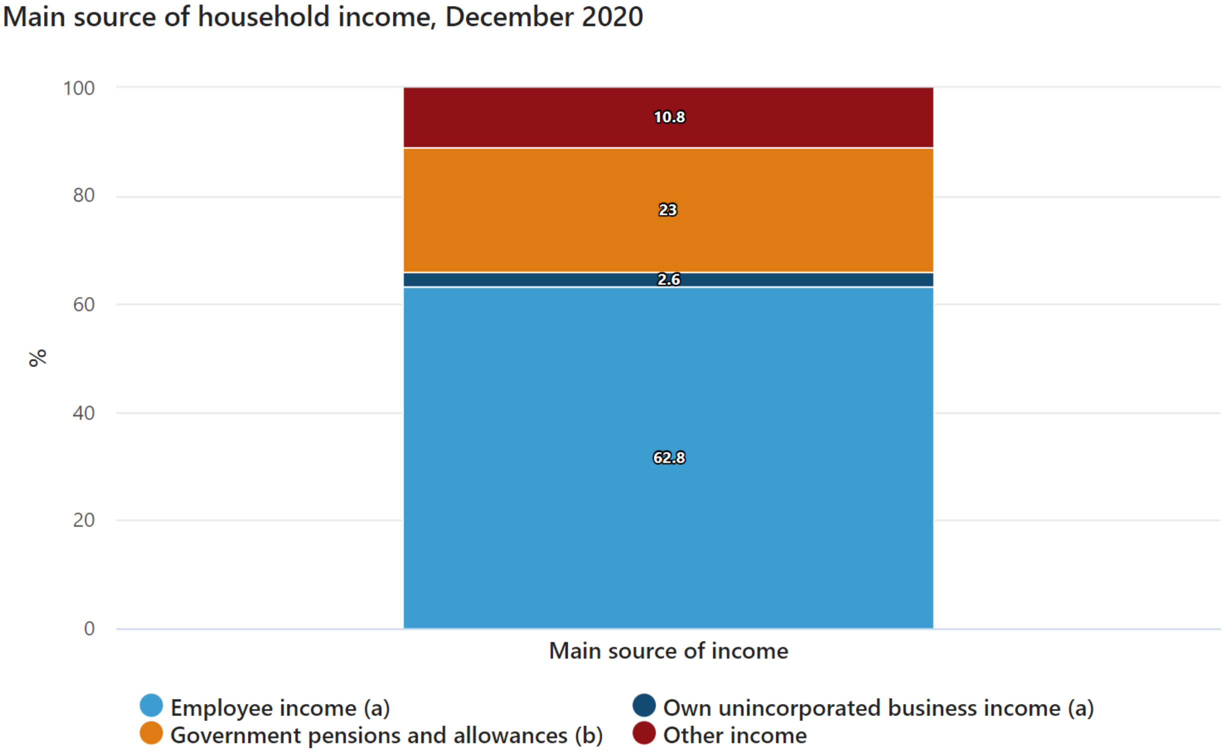
Question 27. (4 marks)



The above graph shows the percentage of the Australian populations source of income for the year 2020. The population of Australia in 2020 was 25 727 066 with 18 230 100 earning an income.

(a) How many of the 18 230 100 people earned an income from Government pensions and allowances? **2**

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(b) What percentage of the population were income earners? **2**

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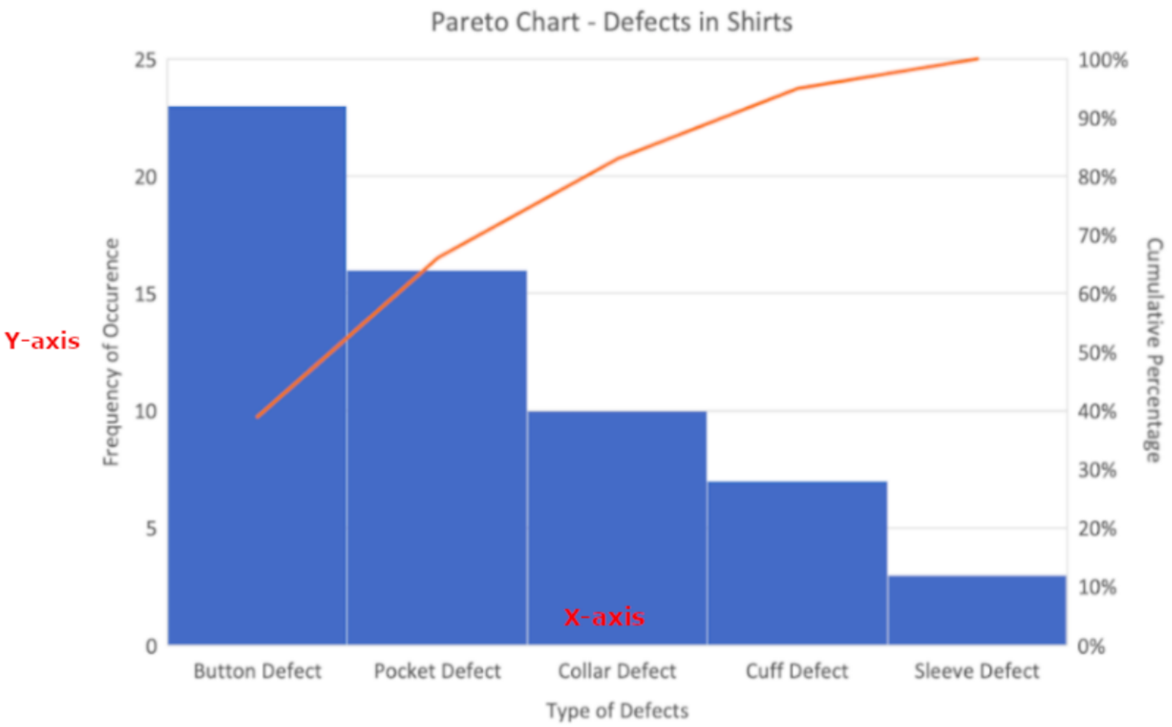
Question 28. (2 marks)

A shirt manufacturer asked customers to fill in a survey asking their main reason for dissatisfaction with their purchase, the feedback was organised into the following Pareto chart.



(a) If the manufacturer addresses the Button and Pocket defect problems, what percentage of customers would have their main reason addressed? **1**

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....

(b) Which additional defect(s) would they need to address to ensure they were meeting the needs of 95% of their customers? **1**

....

....

Question 29. (6 marks)

Georgia deposits \$10000 into a high interest saving account that earns an interest rate of 5% p.a. compounded yearly if she leaves it there for 5 years

(a) Calculate the amount of interest earned at the end of the first year. **1**

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(b) Use the table below to calculate the amount the investment is worth over the 5 years. **2**

Year	Interest	Amount in account
0		\$10000
1		
2		
3		
4		
5		

(c) How much more would she have in the account if the rate of 5% was compounded monthly. **3**

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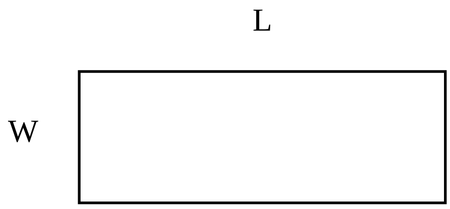
Question 30. (4 marks)

A rectangle has a perimeter of 2m. If 5cm is taken from the length and added to the width, the rectangle becomes a square.

a) Show that the following equations could be used to model the situation. **2**

$$2L + 2W = 200$$

$$L - 5 = W + 5$$



b) Find the side length of the square. **2**

End of Exam

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

REFERENCE SHEET

Measurement

Precision

Absolute error $= \frac{1}{2} \times$ precision

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

Length, area, surface area and volume

$$l = \frac{\theta}{360} \times 2\pi r$$

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(x + y)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$A = \frac{1}{2}ab\sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

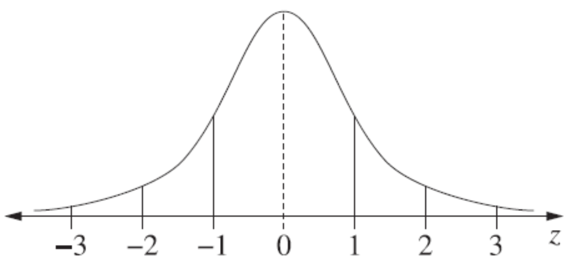
$$S = V_0(1 - r)^n$$

Statistical Analysis

$$z = \frac{x - \bar{x}}{s}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between –1 and 1
- approximately 95% of scores have z-scores between –2 and 2
- approximately 99.7% of scores have z-scores between –3 and 3

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MULTIPLE CHOICE ANSWER SHEET (12 marks)

Name: ……………………………………………….

Student Number: ……………………………………………….

Choose the best answer by filling in the circle corresponding to A, B, C or D.

	A	B	C	D
1	□	□	□	□
2	□	□	□	□
3	□	□	□	□
4	□	□	□	□
5	□	□	□	□
6	□	□	□	□
7	□	□	□	□
8	□	□	□	□
9	□	□	□	□
10	□	□	□	□
11	□	□	□	□
12	□	□	□	□